Title Page

Project Title: Tour and Travel Management System

Developed By:

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Acknowledgement

We would like to express our sincere gratitude to our mentor for their valuable guidance, support, and encouragement throughout the development of this project. Their insights and feedback played a crucial role in the successful completion of our work.

We are also thankful to our faculty members and institution for providing us with the necessary resources and a supportive environment to carry out this project.

Finally, we would like to extend our heartfelt thanks to our friends and family members for their continuous motivation and support during the entire development process.

Abstract

1. Project's Aim

The main aim of this project is to develop a **Tour and Travel Management System** that facilitates customers to plan and book their tours conveniently through an easy-to-use online platform. Instead of manually visiting travel agencies, the system provides a digital solution to book flights, buses, and trains while storing customer and booking details securely in a database.

2. Intended Audience

The project is intended for:

- Individuals planning domestic or international tours.
- Customers who prefer online booking systems over traditional methods.
- Travel agencies who wish to digitalize their services.

3. Cool Features of the System

- **Multiple Travel Options:** Customers can choose from different flights, buses, and trains.
- **Feedback System:** Customers can submit their feedback, which helps in improving service quality.
- **Centralized Database:** All booking and customer information is stored in a secure database for quick access.
- **User-Friendly Interface:** Designed with simple HTML and CSS for ease of use.

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Technology

The **Tour and Travel Management System** has been developed using the following technologies:

1. Server Technologies

- **Apache Tomcat:**
  Tomcat is an open-source web server that supports Java Servlets and JSP. It is lightweight, secure, and widely used for enterprise-level applications.
  Tomcat is highly scalable and can run either as a standalone server or alongside other servers like Microsoft IIS or Netscape Server. It supports session management, concurrency, and high traffic load handling, which is essential for a tour and travel booking application.

- **Oracle Express Edition (XE):**
  Oracle XE is a reliable, secure, and scalable database system. It helps store customer information, booking records, travel packages, and feedback data. The database also provides indexing and query optimization features to ensure fast retrieval of travel and booking details.

2. Software Technologies

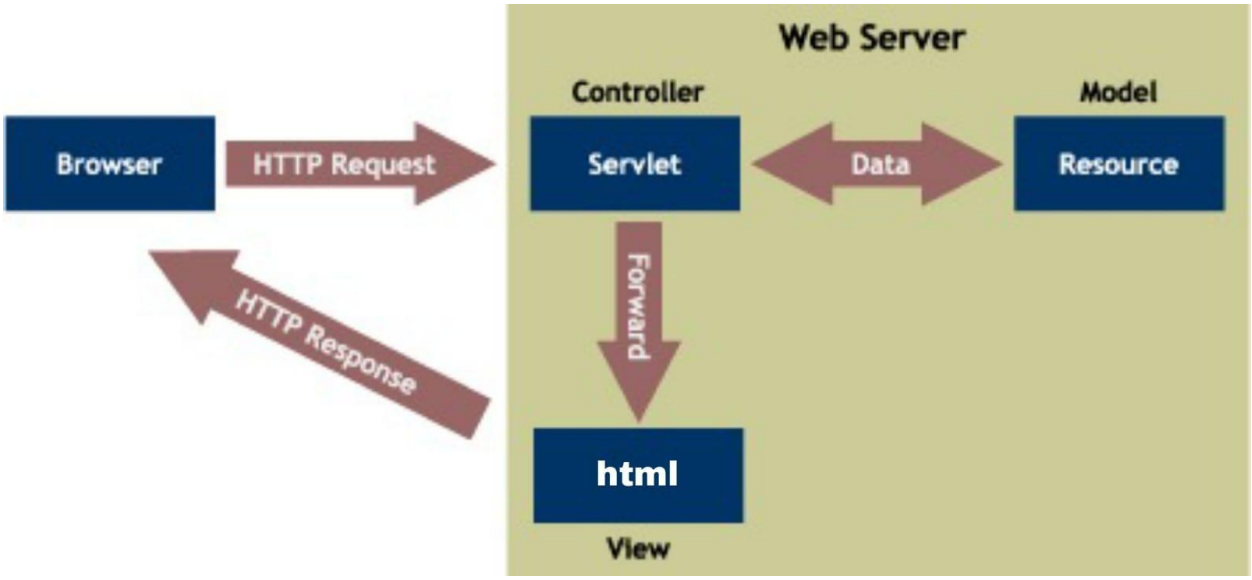
- **Java Servlets:**
  Servlets are Java programs that run on the server and handle requests from clients. In this project, Servlets manage user login, booking confirmation, feedback submission, and database communication. They provide better performance than traditional CGI scripts and are portable across all platforms.

- **JDBC (Java Database Connectivity):**
  JDBC is used to connect Java Servlets with the Oracle database. It enables secure data manipulation such as insertion, deletion, and updates for customer and booking records.

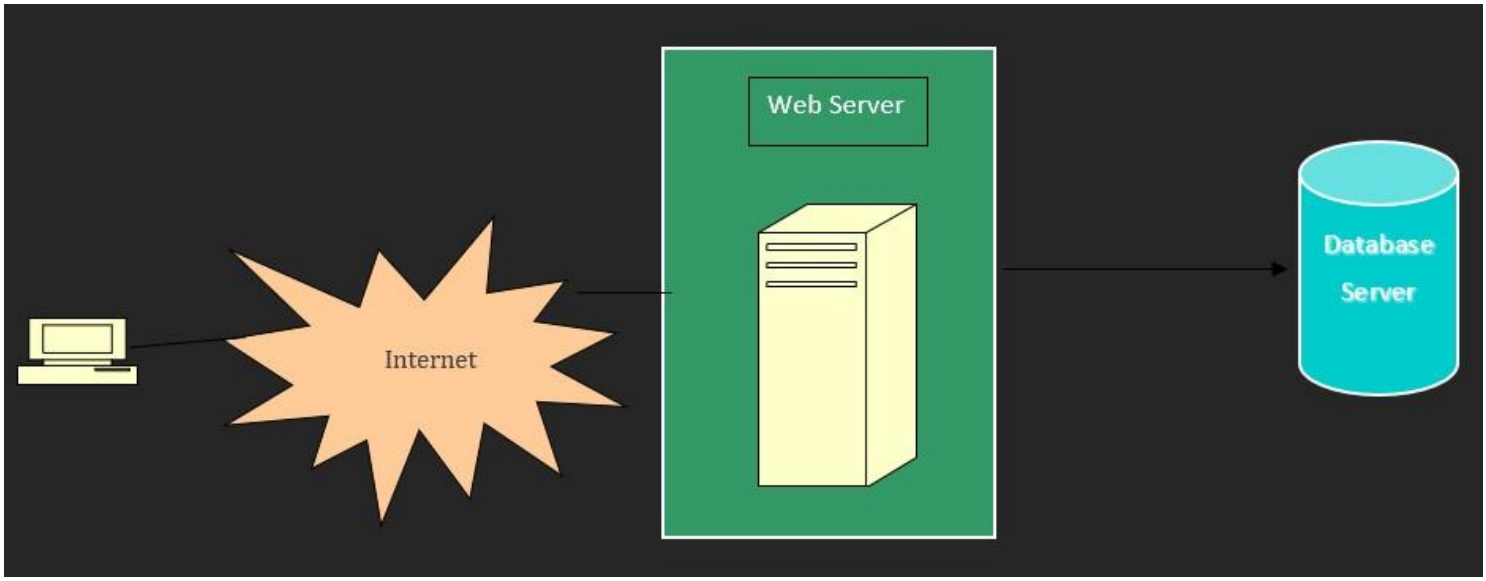
- **HTML & CSS:**
  HTML provides the structure of web pages, while CSS enhances the design and layout. Together, they make the system more user-friendly and attractive.

System Architecture

Model - View – Controller



Application's Overall Architecture:



Feasibility Study

The feasibility study determines whether the project is practical and beneficial for real-world implementation.

1. Economic Feasibility

This project is highly economical because it uses **free and open-source tools** (Tomcat, Java, Oracle XE). No additional hardware is required beyond a basic computer and server environment. It reduces paperwork, manpower, and overall maintenance costs for travel agencies.

2. Technical Feasibility

The project uses **industry-standard technologies** such as Java and Oracle, which are platform-independent, scalable, and secure. The application is designed to handle multiple users simultaneously, making it suitable for small as well as large-scale travel businesses. It can be easily integrated with third-party tools like payment gateways and Google Maps in the future.

3. Behavioural Feasibility

One of the major concerns in system adoption is **user acceptance**. Customers are already familiar with online booking systems (like MakeMyTrip, Yatra, etc.). Hence, they will find this system intuitive and convenient.
Also, travel agents and administrators will require very minimal training to use the system, as it provides simple forms and automated data handling.

Characteristics of the Proposed System

1. **Automation:**
   The proposed system automates the entire booking process, reducing manual effort and chances of human error.

2. **Speed and Accuracy:**
   Searching for travel packages and booking confirmation becomes faster and more accurate compared to manual systems.

3. **Centralized Database:**
   All travel and booking records are maintained in a centralized Oracle database, making it easy to retrieve and update information at any time.

4. **User-Friendly Interface:**
   Designed with HTML & CSS, the system ensures customers with little technical knowledge can use it easily.

5. **Security:**
   Authentication and session handling prevent unauthorized access. Sensitive data like customer details and bookings are secured in the database.

6. **Scalability:**
   The system can be upgraded easily by adding new features such as mobile app integration, payment gateways, or additional services (hotels, cabs, etc.).

7. **Reliability:**
   The use of Java Servlets and Oracle database ensures high reliability, crash protection, and data consistency.

Data Tables

Users Table:

```
Name                                      Null?    Type
----------------------------------------- -------- ---------------
EMAIL                                     NOT NULL VARCHAR2(100)
NAME                                               VARCHAR2(100)
PHONE                                              VARCHAR2(15)
SECURITY_QUESTION                                  VARCHAR2(100)
ANSWER                                             VARCHAR2(100)
PASSWORD                                           VARCHAR2(100)
```

Feedback Table:

```
Name                                      Null?    Type
----------------------------------------- -------- ---------------
ID                                        NOT NULL NUMBER
NAME                                               VARCHAR2(100)
EMAIL                                              VARCHAR2(100)
MESSAGE                                            CLOB
CREATED_AT                                         DATE
```

Flight Booking Table:

```
Name                                      Null?    Type
----------------------------------------- -------- ---------------
BOOKING_ID                                NOT NULL NUMBER
FLIGHT_ID                                          NUMBER
PASSENGER_NAME                                     VARCHAR2(100)
PASSENGER_EMAIL                                    VARCHAR2(100)
PASSENGER_PHONE                                    VARCHAR2(20)
NUM_PASSENGERS                                     NUMBER
TOTAL_AMOUNT                                       NUMBER
PAYMENT_STATUS                                     VARCHAR2(20)
PAYMENT_METHOD                                     VARCHAR2(50)
TRANSACTION_ID                                     VARCHAR2(100)
BOOKED_AT                                          DATE
```

Flight Available Table:

```
Name                                      Null?    Type
----------------------------------------- -------- ---------------
FLIGHT_ID                                 NOT NULL NUMBER
FLIGHT_NAME                                        VARCHAR2(50)
SOURCE                                             VARCHAR2(50)
DESTINATION                                        VARCHAR2(50)
TRAVEL_DATE                                        DATE
PRICE                                              NUMBER
SEATS_AVAILABLE                                    NUMBER(5)
```

Bus Booking Table:

```
Name                                      Null?    Type
----------------------------------------- -------- ---------------
BOOKING_ID                                NOT NULL NUMBER
BUS_ID                                             NUMBER
PASSENGER_NAME                                     VARCHAR2(100)
PASSENGER_EMAIL                                    VARCHAR2(100)
PASSENGER_PHONE                                    VARCHAR2(20)
NUM_PASSENGERS                                     NUMBER
TOTAL_AMOUNT                                       NUMBER
PAYMENT_STATUS                                     VARCHAR2(20)
PAYMENT_METHOD                                     VARCHAR2(50)
TRANSACTION_ID                                     VARCHAR2(100)
BOOKED_AT                                          DATE
```

Bus Available Table:

```
Name                                      Null?    Type
----------------------------------------- -------- ---------------
BUS_ID                                    NOT NULL NUMBER
BUS_NAME                                           VARCHAR2(50)
SOURCE                                             VARCHAR2(50)
DESTINATION                                        VARCHAR2(50)
TRAVEL_DATE                                        DATE
PRICE                                              NUMBER
SEATS_AVAILABLE                                    NUMBER(5)
```

Train Booking Table:

```
Name                                      Null?    Type
----------------------------------------- -------- ---------------
BOOKING_ID                                NOT NULL NUMBER
TRAIN_ID                                           NUMBER
PASSENGER_NAME                                     VARCHAR2(100)
PASSENGER_EMAIL                                    VARCHAR2(100)
PASSENGER_PHONE                                    VARCHAR2(20)
NUM_PASSENGERS                                     NUMBER
TOTAL_AMOUNT                                       NUMBER
PAYMENT_STATUS                                     VARCHAR2(20)
PAYMENT_METHOD                                     VARCHAR2(50)
TRANSACTION_ID                                     VARCHAR2(100)
BOOKED_AT                                          DATE
```

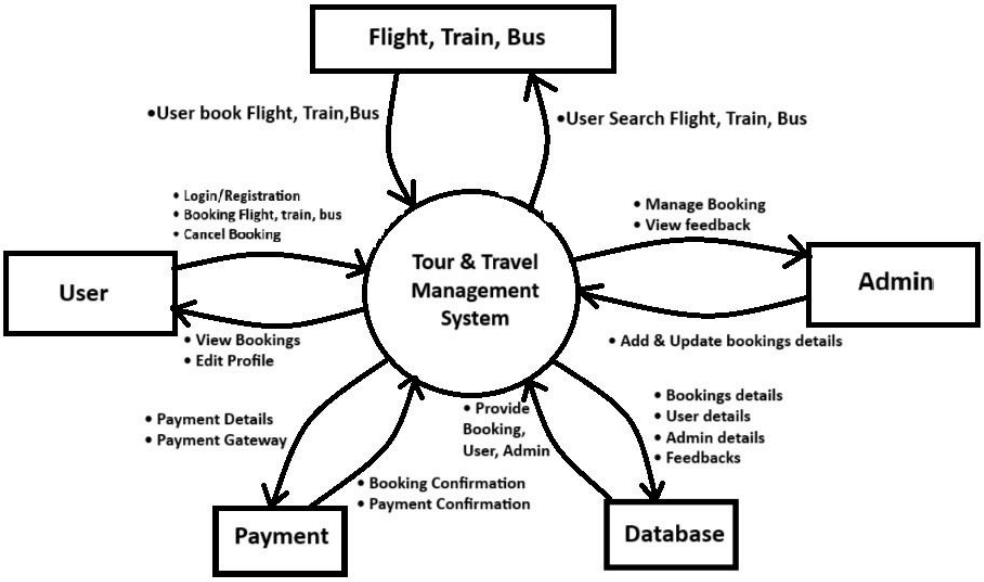
Train Available Table:

```
Name                                      Null?    Type
----------------------------------------- -------- --------------
TRAIN_ID                                  NOT NULL NUMBER
TRAIN_NAME                                         VARCHAR2(50)
SOURCE                                             VARCHAR2(50)
DESTINATION                                        VARCHAR2(50)
TRAVEL_DATE                                        DATE
PRICE                                              NUMBER
SEATS_AVAILABLE                                    NUMBER(5)
```

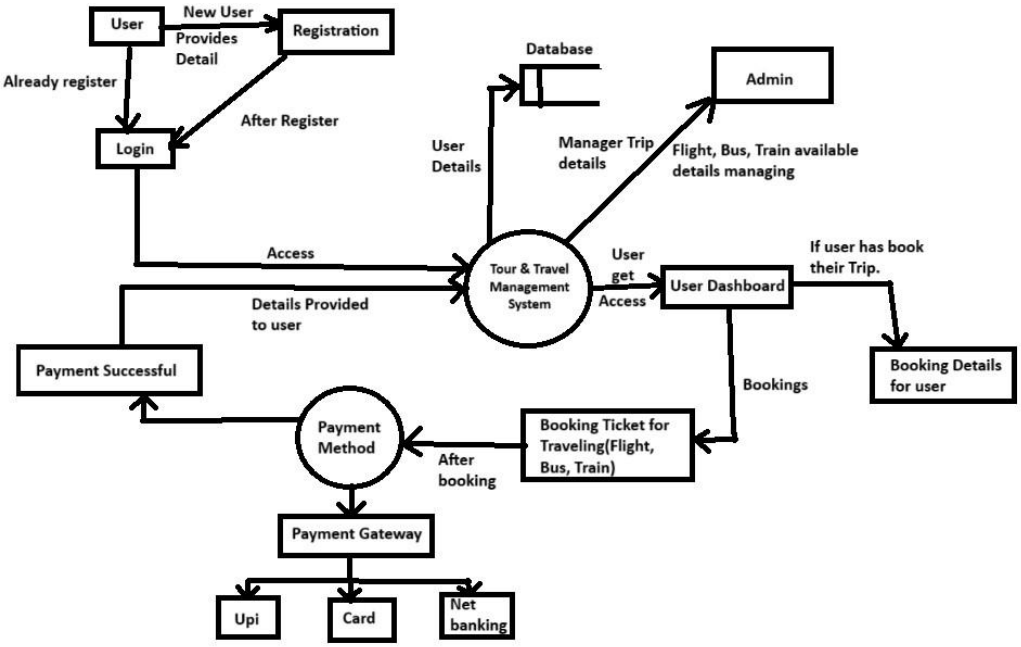
Admin Table:

```
Name                                      Null?    Type
----------------------------------------- -------- ---------------
ADMIN_ID                                  NOT NULL VARCHAR2(50)
ADMIN_PASS                                NOT NULL VARCHAR2(50)
```

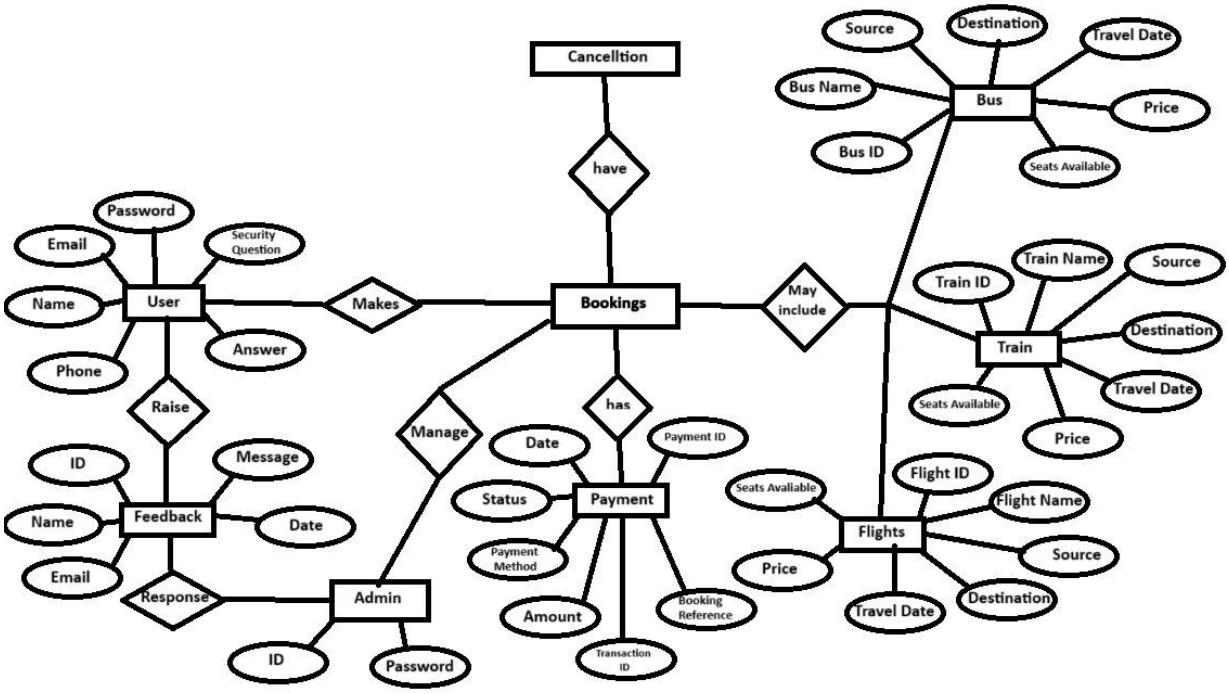
0-Level DFD



1st-Level DFD



ER Diagram


Cancelltion
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Bookings
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User
Password
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Security Question
Name
Answer
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Raise
Feedback
ID
Message
Name
Date
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Admin
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Manage
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Payment
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Bus Name
Price
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Seats Available
Train
Train ID
Train Name
Source
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Source
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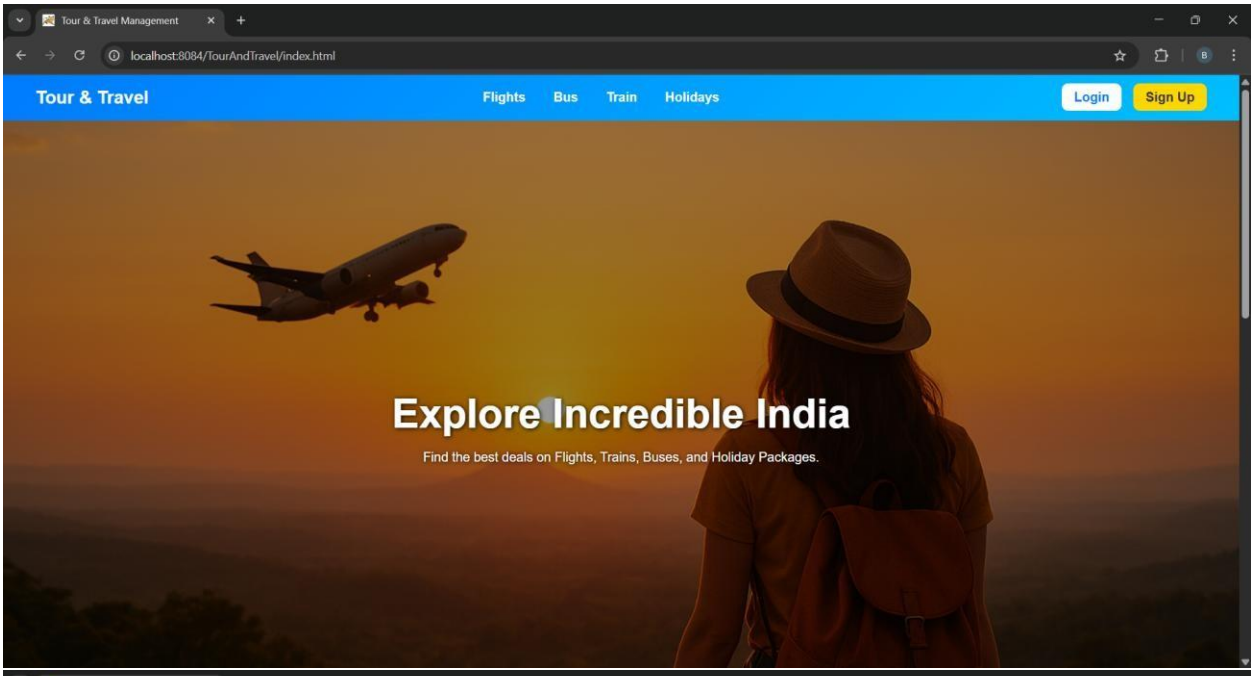
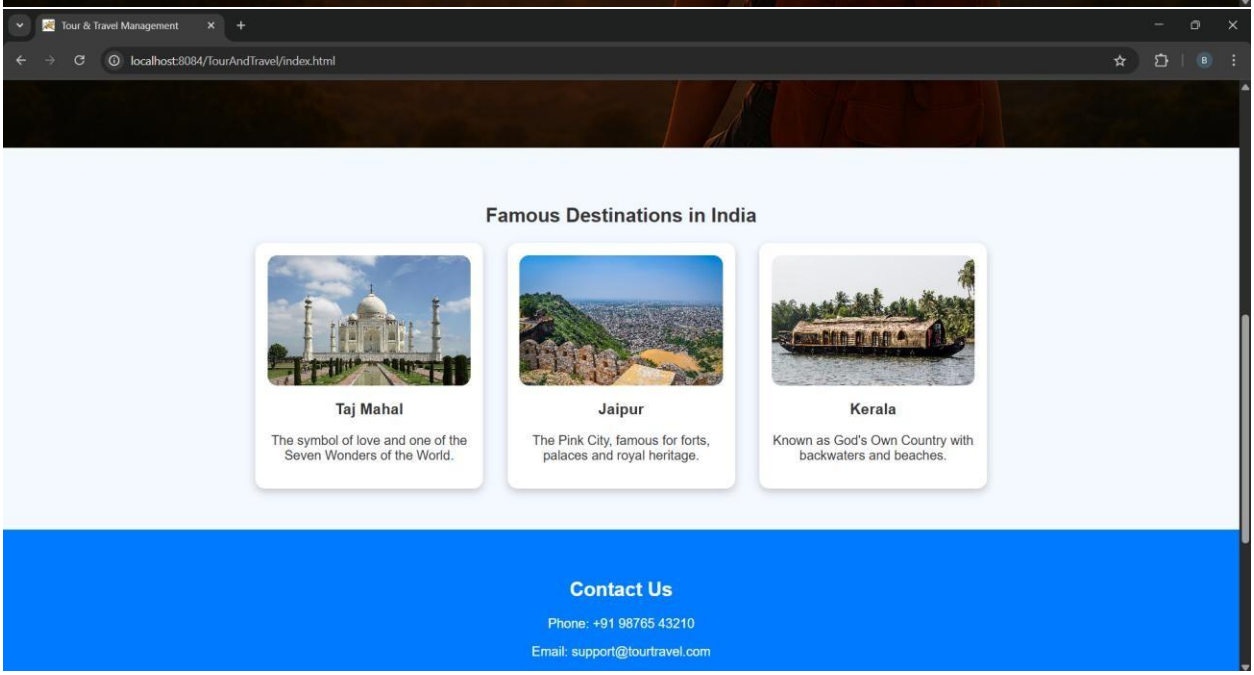
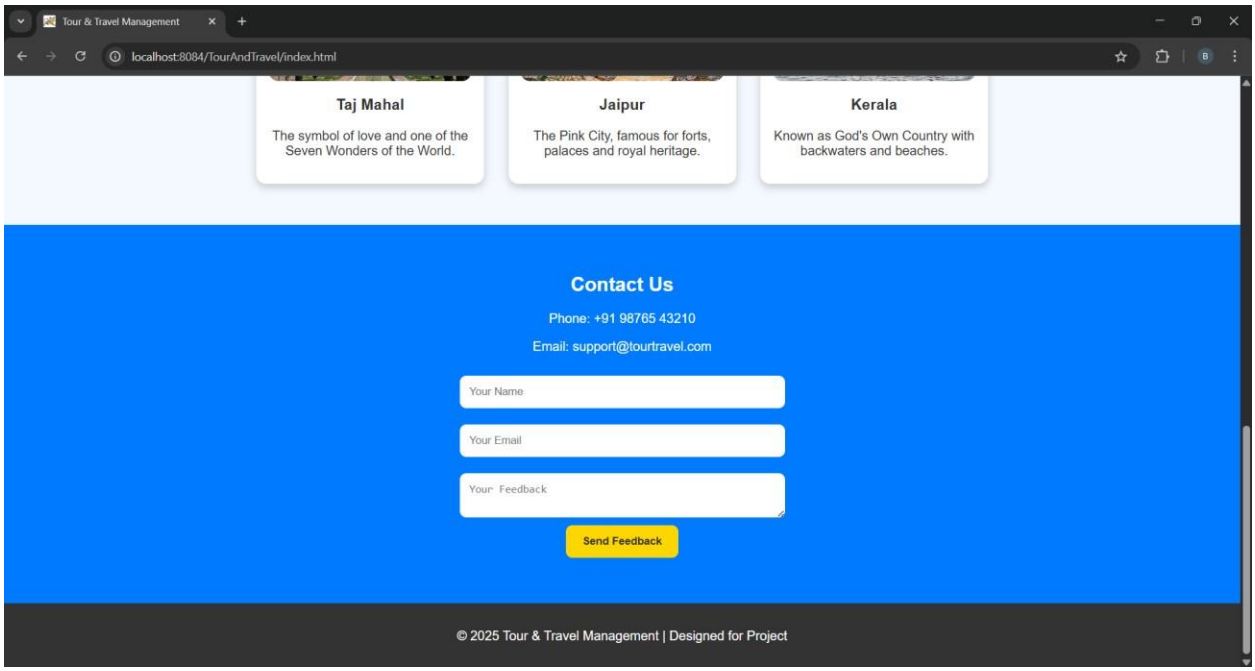
Travel Date
Destination

Screenshots

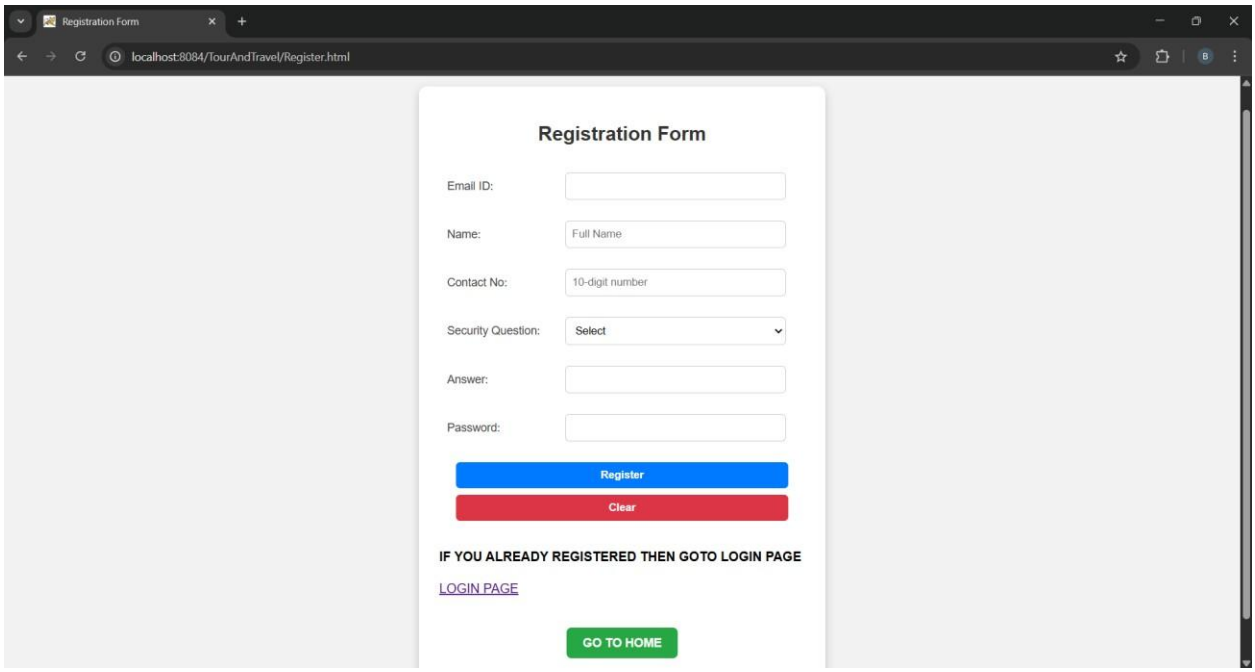
Home Pages:



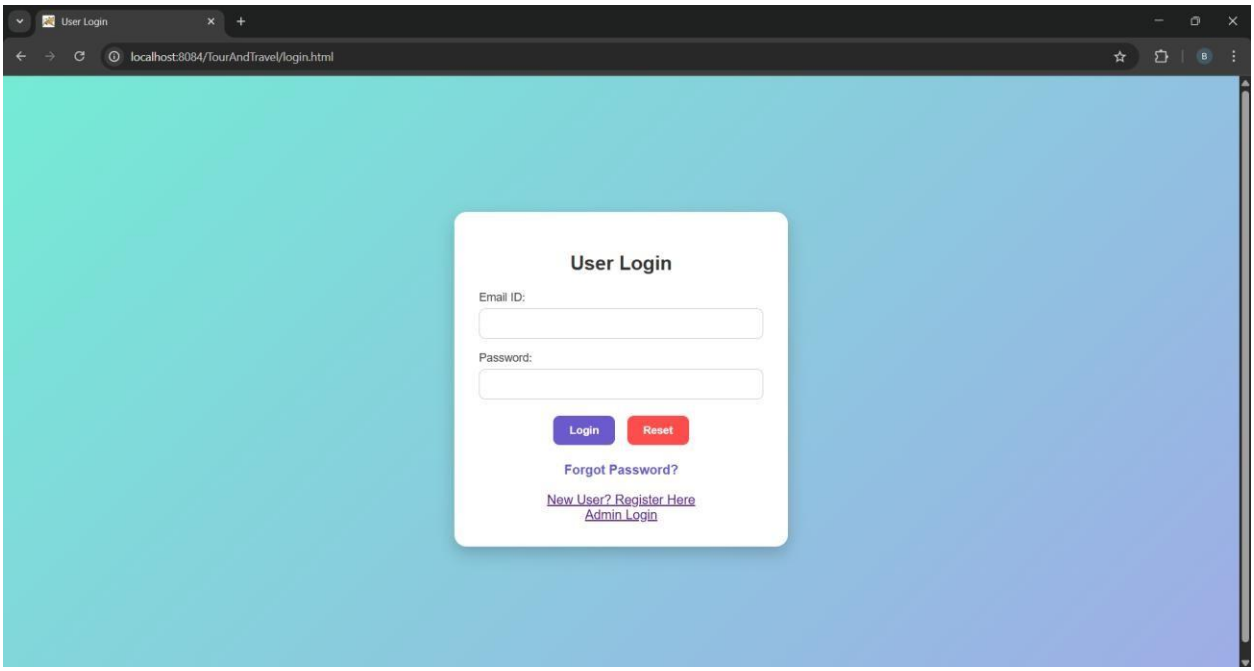




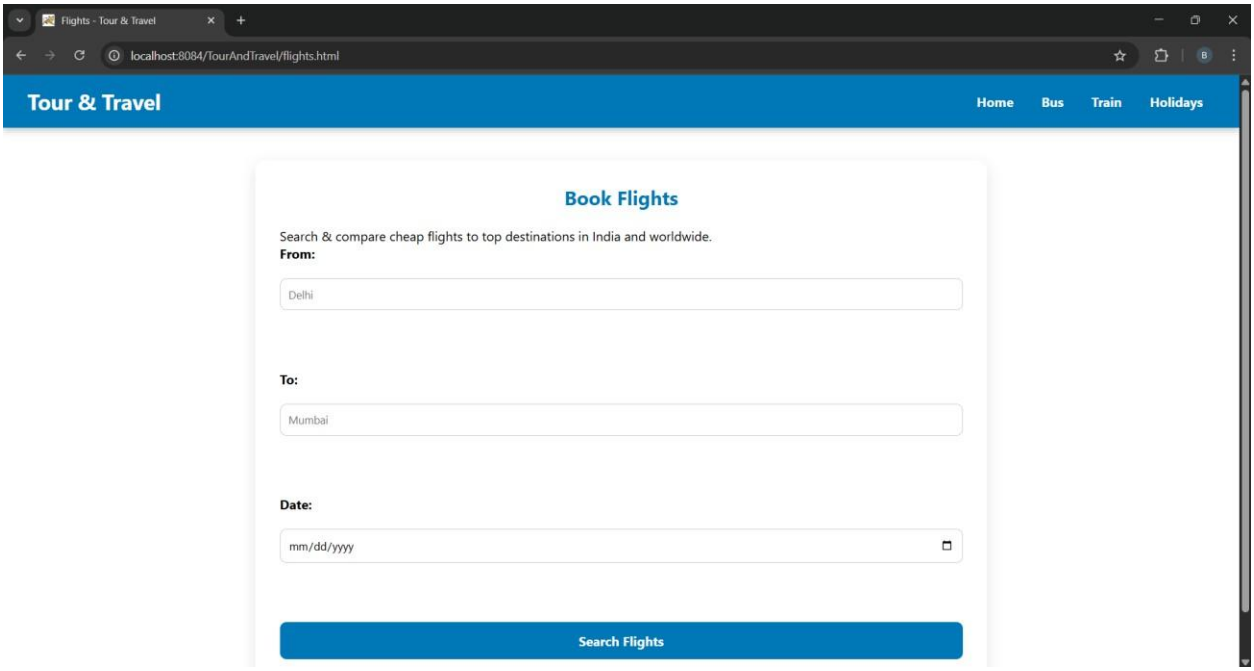
Registration Page:



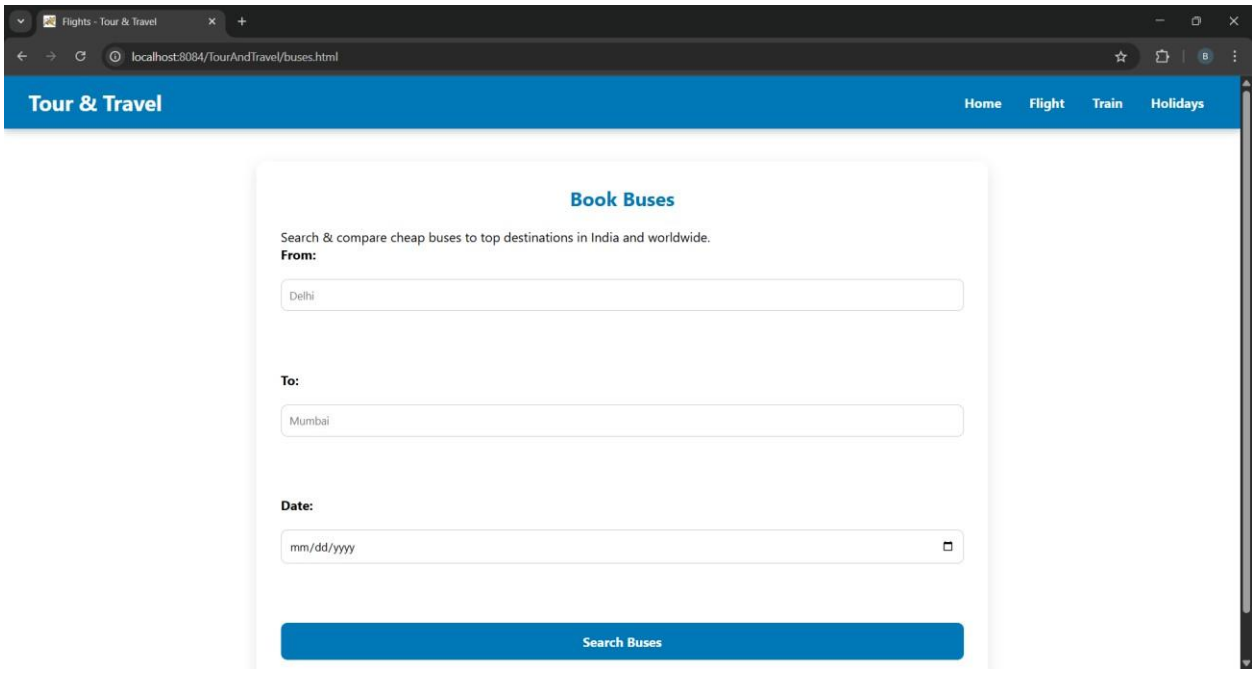
Login Page:



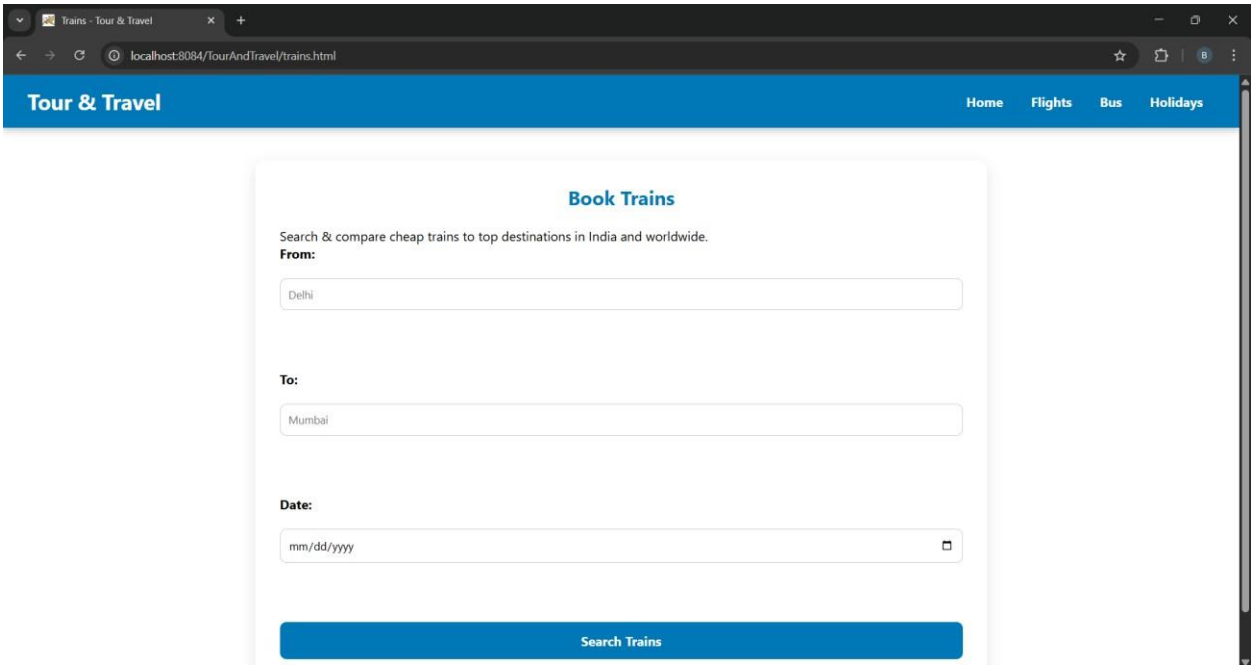
Flights Search page:



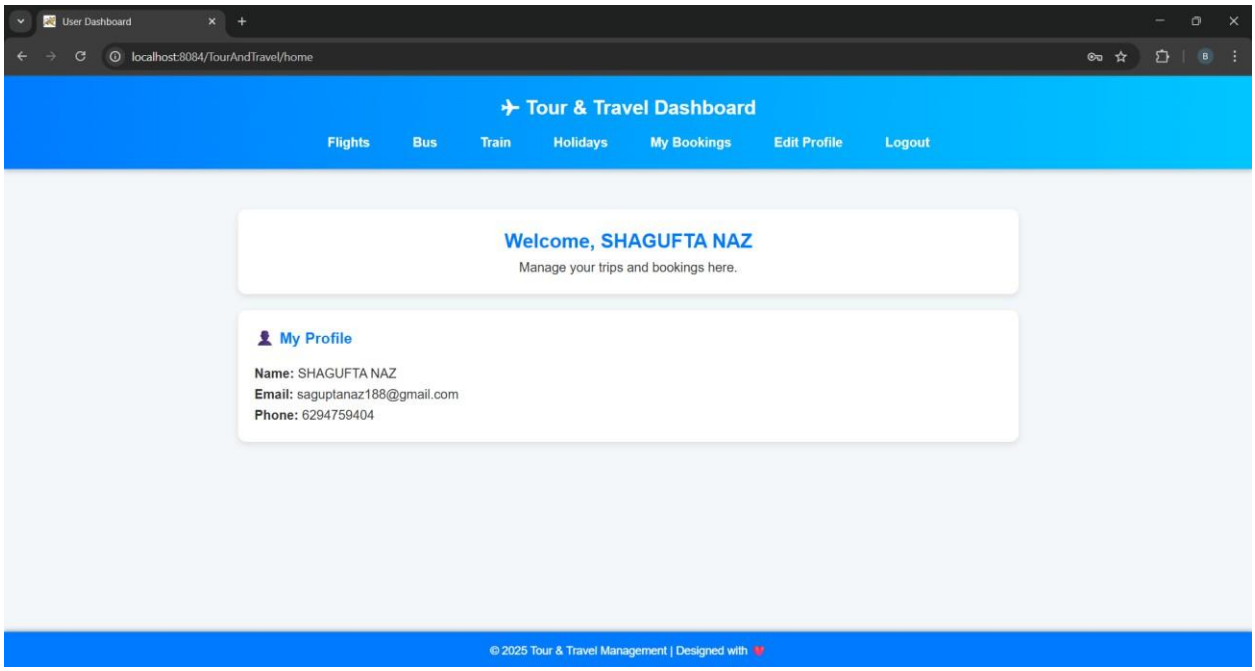
Bus Search Page:



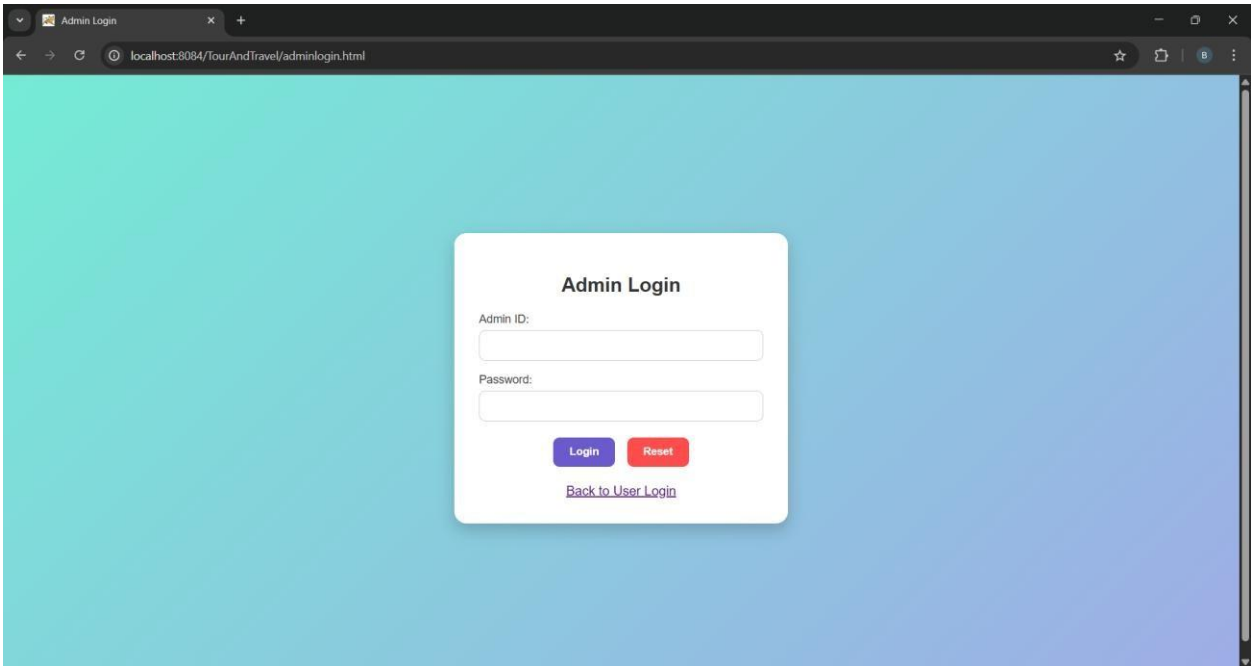
Train Search Page:



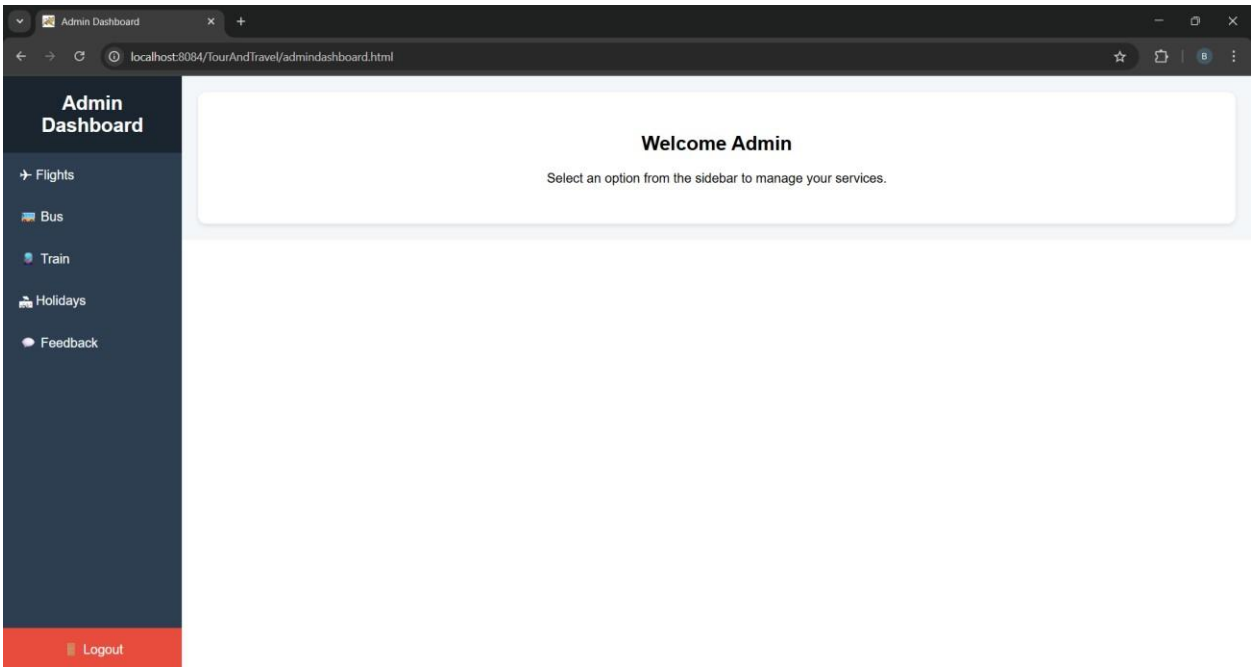
User Dashboard Page:



Admin Login Page:



Admin Dashboard Page:



Future Scope

The Tour and Travel Management System can be extended in several directions:

1. **Payment Gateway Integration:**
   Implementing secure online payment options (UPI, credit/debit card, net banking) for real-time transaction processing.

2. **Mobile Application:**
   Development of Android and iOS mobile apps so customers can book travel services on the go.

3. **AI-based Recommendations:**
   The system can provide smart suggestions for travel packages based on user history, preferences, and budget.

4. **Multi-Language & Multi-Currency Support:**
   Expanding the system for international use by supporting multiple languages and currencies.

5. **Hotel & Cab Booking Integration:**
   The system can be extended to provide a one-stop solution for customers (hotel booking, cab rental, sightseeing packages).

6. **GPS & Google Maps Integration:**
   Providing real-time travel tracking, location mapping, and distance/time calculation for better customer experience.

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